Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Assessment Task No 1			
Topic:	**Introduction to Machine Learning**	**Week No.**	**2-3**
Course Code:	**CSST102**	**Term:**	**1st Semester**
Course Title:	**Basic Machine Learning**	**Academic Year:**	**2026-2027**
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Performance Task 1: From Data to Prediction — Analyzing Real-World Machine Learning Applications

Part A — Identify the ML Problem

Question	Your Answer
1. What is the application?	Product recommendation
2. What problem are you trying to solve?	Look at what a customer has bought or viewed, and suggest other products they likely want.
3. What are the inputs?	A customer past purchases, items viewed and items added to cart.
4. What is the expected output?	A List of recommended products for that customer.
5. What patterns or relationships might the ML system learn?	Customers who buy similar things tend to like similar products. Like if someone who buys a phone often also buys a phone case.
6. What data would the system need?	Records of many customer purchases and browsing history.

Part B — Traditional Programming vs Machine Learning

Traditional Programming

The programmer would need to create many rules explaining how different user preferences should be matched with products. For example, the system could recommend products based on category, price range, brand, or rating.

Machine Learning

Instead of writing rules, we give the computer records of what many customers bought and viewed before. The computer looks at all these examples and learns on its own which products are often bought together or liked by similar customers. Then it can recommend products to a new customer baked in what they bought.

Part C — Create Your Own Mini Dataset

Input (bought/ viewed/ add to cart)	Output (Recommended)
1. Bought a phone	Phone case
2. Bought a Laptop	Laptop bag
3. Viewed a gaming-related products	Gaming chair
4. Add to cart a camera	Camera strap
5. Add to cart some coffee maker	Coffee beans

1. Identify the pattern.

Each purchased or viewed item is paired with a related item that people are usually interested in.

2. Determine the missing output.

If a customer views several phone-related products, the system could recommend a phone case.

3. Write the mathematical relationship/function.

f(Customer's Viewed Products, Purchase History) = recommended product

4. Explain how a machine learning algorithm could learn this relationship from the examples

The ML algorithm would study many examples of customers viewed or purchased products and the products they later bought. By finding patterns between them, the algorithm can learn to predict which products a new customer might be interested in.

Part D — Explain the Training Process

How does a Machine Learning algorithm learn from data?

A machine learning algorithm uses input data, like the products a customer views or buys, and an output, like a recommended product. During training, it looks at many examples and finds patterns. It learns the relationship between what customers like and what products they might want. After training, it can make a prediction and recommend products to new customers.

Reflection:

Why is learning from data useful instead of manually programming every possible rule?

Learning from data is useful because there are too many possible customer preferences to program manually. Machine learning can find patterns in the data and make recommendations based on what customers usually like. It can also improve as it gets more data.